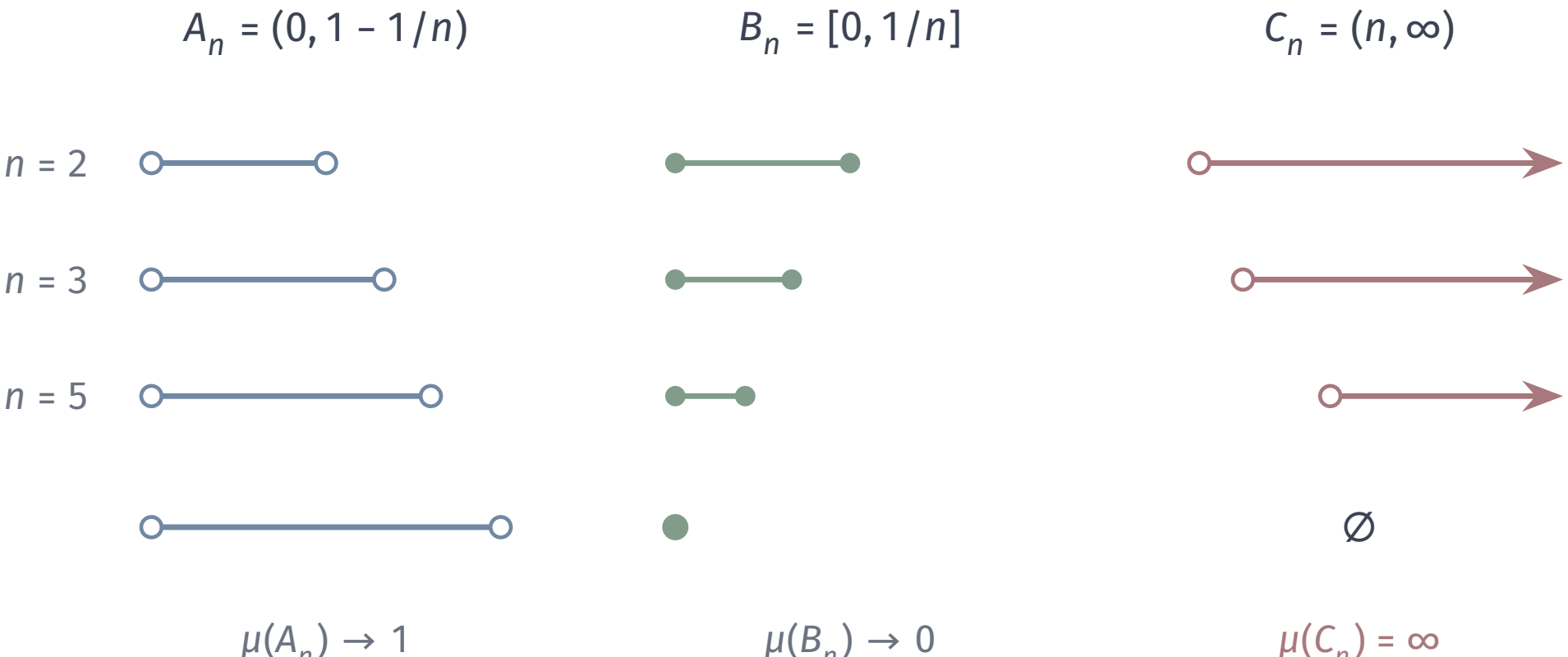
$A_n = (0, 1 - 1/n)$
$B_n = [0, 1/n]$
$C_n = (n, \infty)$
$n = 2$
$n = 3$
$n = 5$
$\varnothing$
$\mu(A_n) \to 1$
$\mu(B_n) \to 0$
$\mu(C_n) = \infty$